ROEHAMPTON AMATEUR RADIO SOCIETY

Your first steps in amateur radio

Week 1 · UK Foundation preparation

A 90-minute lesson with a learner guide, classroom activities, original diagrams and a knowledge check.

How to use this pack

Start with the learner guide. Complete the worksheet and quiz before reading the answers. Tutor notes and source checks follow the learner materials.

- Learner guide
- Worksheet
- Quiz
- Explained answers
- Tutor notes
- Sources and scope

Bring the official reference booklet

The chart and table activities use EX307, available through rsgb.org/main/clubs-training/forms/. The official booklet is linked separately, not reproduced here.

Checked 6 October 2026 against the RSGB March 2026 syllabus revision. This independent RARS teaching draft introduces part of the syllabus; it is not a full exam-preparation course or an official RSGB, Ofcom or Essex Ham publication.

Week 1 learner guide

1. A radio conversation starts with a few choices

Imagine two RARS members want to speak to each other from different places around campus. Before touching the transmit button, they need to agree where to listen and how to communicate. They also need permission to transmit there.

Three words will help you make sense of almost every radio conversation:

Word	Meaning	Example
Frequency	How quickly a signal repeats	A display reading 145.500 MHz
Band	A range of frequencies	The amateur 2-metre band
Mode	How the information is carried	FM voice

The display example is for learning vocabulary. Choosing an actual channel also involves your licence, the current band plan and listening for existing users. We will practise those operating decisions later.

2. What is amateur radio?

Amateur radio, often called **ham radio**, lets people learn about radio by using it. You might build an antenna, compare signals, speak to another operator or exchange information between computers over a radio link.

The amateur licence is for personal learning, technical investigation and leisure. It is not a business radio licence: business traffic and commercial advertising are outside its purpose. [Ofcom source: S4 in the source notes.]

You do not need to arrive as an electronics expert. Foundation is the starting level. The usual progression is **Foundation, Intermediate, Full**, with further study building on the earlier level.

People enjoy different parts of the hobby. A **net** is an organised on-air group conversation. A **contest** is an activity with rules for making and recording contacts. **DX** refers to distant contacts. A **QSL** is a confirmation of a contact, which may be a card or an electronic record. These words are useful background; the activities are not compulsory parts of being a radio amateur.

Passing the exam and obtaining a licence are separate steps. Follow the RSGB result process, then apply to Ofcom for your licence. You may also experience transmitting during training under permitted direct supervision; the licensed supervisor remains responsible. In this first session, our activities work without transmitting. [S2, S4]

Pause and explain: why is testing an antenna for personal learning different from using the same radio to dispatch paid delivery jobs?

3. Radio spectrum: shared space

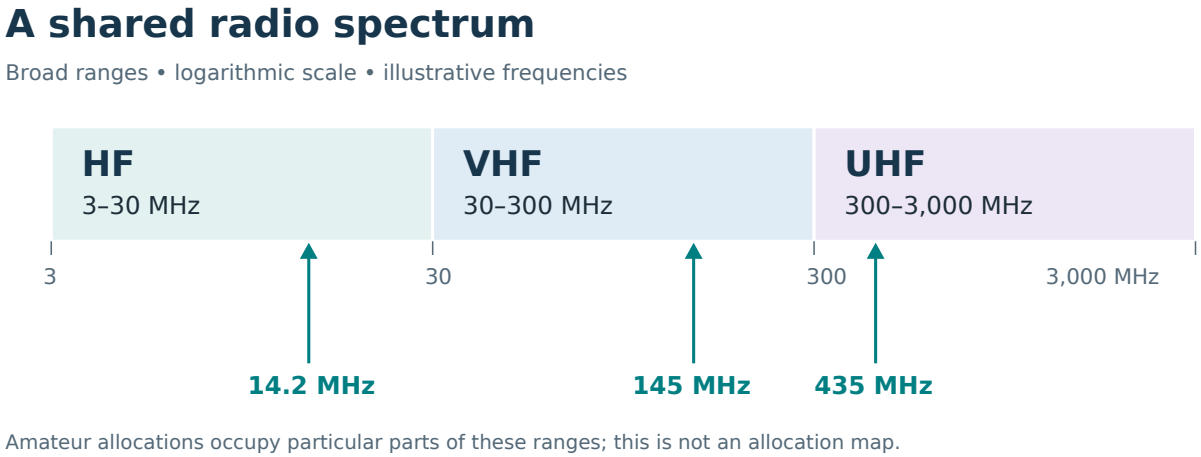
The **radio spectrum** is the range of radio frequencies. Broadcast stations, navigation systems, mobile networks, satellites and many other services use parts of it. Amateur radio has particular allocations within that larger spectrum.

Being able to tune a radio to a frequency does not mean you may transmit on it. Think of the radio's tuning range as its technical capability; the licence describes your permission.

Three ranges to remember

Range	Full name	Frequency range
HF	High frequency	3–30 MHz
VHF	Very high frequency	30–300 MHz
UHF	Ultra high frequency	300–3,000 MHz

Remember the boundaries as a sequence: **3, 30, 300, 3,000 MHz**. Each range spans a factor of ten. These are broad spectrum classifications, not continuous amateur allocations.



Worked examples: 7.1 MHz falls in HF. 145 MHz falls in VHF. 435 MHz falls in UHF. For each one, compare the number with the boundaries; do not guess from the number of digits.

Check: 2.4 GHz equals 2,400 MHz. Which of the three ranges contains it?

Allocations and sharing

An **allocation** assigns spectrum to a radio service. The familiar UK amateur 2-metre allocation is **144–146 MHz**, a small part of VHF. A licence table may divide a familiar band into several rows because conditions differ across it.

Primary and **secondary** describe a service's status. A primary allocation is not necessarily exclusive. A secondary service must avoid harmful interference to primary services and cannot demand protection from their legitimate operation. Amateur stations must avoid undue interference in all cases. Always read additional conditions in the relevant row. [S2, S4]

For a first lookup, open the Foundation reference booklet and find the 144–146 MHz row. Read across the row rather than stopping at its frequency column. Notice how status and power appear separately.

4. Frequency counts; wavelength measures distance

Radio waves are electromagnetic waves. A smooth wave drawing is a way to show a repeating pattern, not a picture of a tiny object travelling up and down through the air.

Frequency is the number of complete cycles in one second. Its unit is the **hertz**, written **Hz**. A pattern completing four cycles in one second has a frequency of 4 Hz. That is a slow classroom illustration, not a typical amateur transmission.

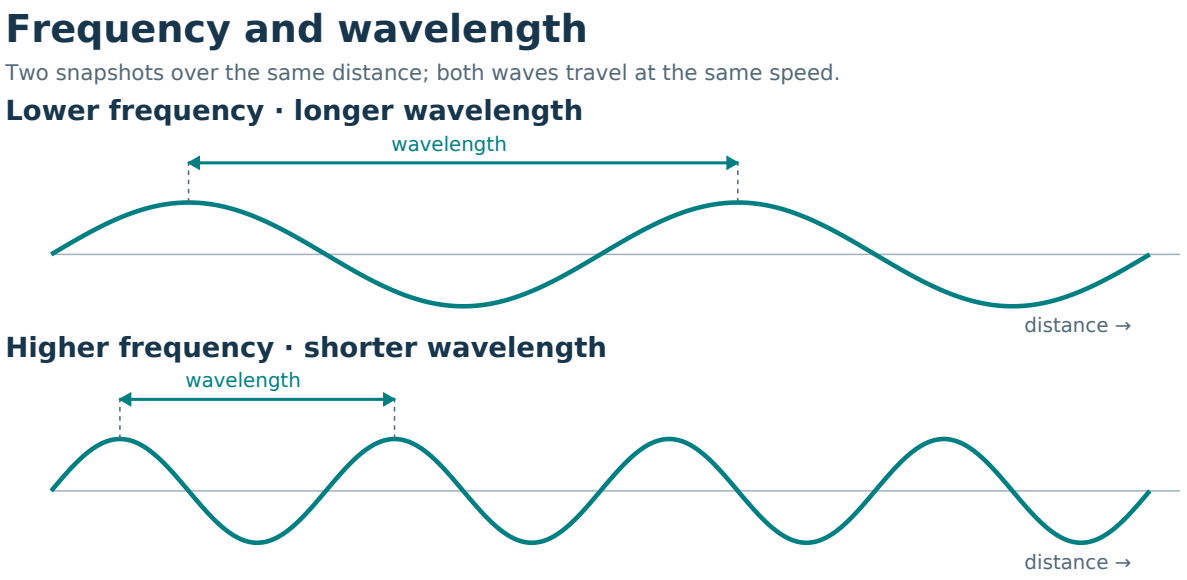
Unit	Meaning
1 Hz	1 cycle per second
1 kHz	1,000 Hz

Unit	Meaning
1 MHz	1,000,000 Hz
1 GHz	1,000,000,000 Hz

For example, 145 MHz means 145 million cycles per second. Hertz already includes “per second”; do not say “hertz per second” when defining a frequency.

Wavelength is the distance covered by one complete spatial cycle. We write it as **λ**, the Greek letter lambda, and measure it in **metres (m)**. On a snapshot of a wave plotted against distance, you can measure from one crest to the next crest.

The axis matters. Crest-to-crest on a **distance** graph measures wavelength. Crest-to-crest on a **time** graph measures the duration of a cycle, called its period. Do not use a time axis to measure a length.



Crest-to-crest on a distance axis measures wavelength.

These are schematic field patterns, not the path followed by a particle.

The relationship

Radio waves in free space travel at the speed of light. With the speed fixed, more cycles each second means each cycle occupies less distance. Therefore:

Higher frequency → shorter wavelength. Lower frequency → longer wavelength.

If frequency doubles, wavelength halves. This does not mean that the higher-frequency wave travels faster or automatically reaches farther.

Why call it “2 metres”?

Amateur band names are convenient approximations. At 145 MHz the wavelength is about **2.07 m**, so that frequency belongs to the band commonly called 2 metres. A band name is neither an exact wavelength for every frequency in the band nor a substitute for its permitted frequency limits.

Similarly, the amateur “20-metre band” lies around 14 MHz. The name does not mean that its waves are exactly 20.000 metres long.

Practise with the official chart

Open the frequency–wavelength chart on page 4 of the current EX307 booklet. The axes use logarithmic spacing: equal-looking gaps do not always mean equal numerical differences.

1. Find a frequency on the horizontal axis, paying attention to units.
2. Move vertically to the diagonal conversion line.
3. Move horizontally to the wavelength axis.
4. Read an approximate value and include its unit.

Reverse these steps to start from wavelength. A frequency near 145 MHz should give a wavelength near 2 m; about 14 MHz gives a little over 20 m. Explain the direction first, then make the estimate. [S2]

Optional numerical check

You can also check a result using **wavelength in metres ≈ 300 ÷ frequency in MHz**. This is an optional aid in this module. The Foundation learning point focuses on understanding the relationship and units; practise the supplied chart rather than treating formula manipulation as an extra Foundation requirement. [S1]

Frequency	Calculation	Approximate wavelength
14.2 MHz	300 ÷ 14.2	21.1 m
145 MHz	300 ÷ 145	2.07 m
435 MHz	300 ÷ 435	0.69 m, or 69 cm

The formula is about physics, not permission. For example, exactly 20 m corresponds to approximately 15 MHz, which is **not** the UK amateur 20-metre allocation. Always use the licence table to decide where operation is allowed.

5. How information travels: modes and modulation

A transmitter creates a radio-frequency signal. To communicate, we arrange for that signal to carry information. **Modulation** changes a property of the signal in a controlled way. A receiver recovers the information so that we can hear, read or view it.

Changing **frequency setting** means choosing where the signal sits in the spectrum. Choosing a **mode** means choosing how to convey information there. They are different decisions.

Term	First explanation	What you might encounter
AM — amplitude modulation	The signal's amplitude varies with the information	Analogue voice
FM — frequency modulation	The instantaneous frequency varies around the carrier frequency	Analogue voice, including many local contacts
SSB — single sideband	A form of AM using one sideband with the carrier suppressed	Analogue voice, often on HF
CW — continuous wave	A carrier is keyed on and off to form Morse characters	Morse communication
Digital voice	Speech is represented digitally for transmission	Voice that needs a compatible digital system
Data mode	Encoded information is exchanged by suitable equipment/software	FT8 is one example

Term	First explanation	What you might encounter
SSTV — slow-scan television	Pictures are sent relatively slowly over radio	Still-image exchanges

This table mixes modulation names with wider operating-mode labels because that is how you will meet the terms on radios and in conversation. Later lessons explain transmitter blocks, waveforms and bandwidth in more detail.

FT8 is designed for short, structured exchanges; it is not simply a free-form keyboard chat. SSTV normally sends still pictures, rather than live broadcast-style moving television. A shared frequency alone is not enough for a useful contact: the stations also need compatible modes and settings.

Try explaining it: if two radios are on the same frequency but using incompatible modes, why might the operators hear noise or fail to decode each other?

6. Know what to remember and what to look up

Remember and explain	Practise looking up
Amateur radio's non-commercial purpose	Allowed frequency limits for the relevant licence
Foundation, Intermediate and Full	Status, power and special conditions in a table row
HF/VHF/UHF boundaries	Frequency–wavelength estimates on the official chart
Frequency in Hz; wavelength in metres	The meaning of the material supplied for the exam
Increasing frequency shortens wavelength	Current references before an operating activity
Frequency, band and mode are different ideas	Later: the appropriate portion of the current operating band plan

A **licence schedule** sets permissions and conditions. A **band plan** helps organise activities within a band. The reference booklet contains examination material; the specimen exam band plan should not replace the current operating band plan for real contacts. [S1]

7. Before next week

Complete the worksheet and the 12-question quiz. Mark your answers using the explained key only after you have tried them yourself. For each mistake, write one sentence explaining the corrected idea.

Then prepare a 60-second answer to: **“What would you like to try with amateur radio, and what would you need to learn first?”**

You are ready to build on this week when you can classify a frequency, explain the wavelength relationship and find a table row without confusing an allocation with the radio's tuning range. Ask for more practice wherever you are unsure.

RARS teaching draft v0.1 · Checked 6 October 2026. Sources and scope · Worksheet · Quiz

Week 1 classroom worksheet

Name: _____ **Date:** _____

Use the learner guide for Activities A–D and the official **EX307 Foundation reference booklet** for the chart and table tasks. Get the booklet from the RSGB exam forms page. These activities do not require transmission.

Activity A · What is the purpose?

For each example, explain whether it fits amateur radio's non-commercial purpose. Assume that suitable licensing and other operating conditions would also be needed.

1. Two learners compare homemade antennas to understand their performance.
2. A delivery company uses amateur channels to allocate its paid jobs.
3. Club members exchange observations about reception conditions.

Explain your reasoning:

_____

_____

Activity B · Sort the spectrum

Classify each frequency as HF, VHF or UHF. These are classification examples, not suggested channels or permission to transmit.

Frequency	Convert to MHz if needed	HF, VHF or UHF?
7.1 MHz		
14.2 MHz		
100 MHz		
145 MHz		
435 MHz		
2.4 GHz		

Does “VHF” mean all those frequencies are allocated to amateurs? Explain.

_____

Activity C · Read a wave and use a chart

1. A repeating signal completes six whole cycles in one second. Its frequency is _____ Hz.
2. On a graph with a **distance** axis, what does crest-to-crest separation measure?
   _____
3. If frequency doubles while wave speed stays constant, wavelength _____.
4. Use the EX307 frequency–wavelength chart to estimate the wavelengths near 14 MHz and 145 MHz. Write the units.

Frequency	Estimated wavelength
About 14 MHz	
About 145 MHz	

5. Start at about 2 m on the wavelength axis and work back to frequency. Your approximate answer: _____.
6. Why does the name “20-metre band” not mean that every frequency in it has exactly a 20 m wavelength?

_____

Optional calculation: use wavelength (m) ≈ 300 ÷ frequency (MHz) to check 145 MHz. Do not replace the chart activity with the calculator.

Activity D · Match the ideas

Write the matching term next to each description. Choose from **FM, SSB, CW, digital voice, SSTV**.

Description	Term
A carrier keyed on and off for Morse	
Speech represented digitally	
Relatively slow transmission of a still picture	
Instantaneous frequency varied by the information	
One sideband transmitted with the carrier suppressed	

A radio changes from FM to a digital voice mode while the displayed frequency stays the same. Has its **frequency**, **mode** or **band** changed? _____

Activity E · Read across a licence-table row

Open the Foundation Licence Parameters table in EX307. For the **144–146 MHz** row, find:

Item	Your answer
Lower and upper frequency limits	
Status of the amateur service	
Maximum peak envelope power for ordinary terrestrial operation	

Now find **10,100–10,150 kHz**. What is the amateur service's status there?

_____

Why is knowing the frequency alone insufficient when deciding whether and how to transmit?

_____

Exit note

One idea I can explain: _____

One question I still have: _____

Keep the completed worksheet for revision. Answer key

Week 1 knowledge check

12 original RARS practice questions · approximately 8 minutes

Choose one answer for each question. Use EX307 for Questions 8 and 12; try the others from memory. This is a weekly learning check, not an official examination paper or a full-syllabus mock.

Questions

1. Which proposed activity fits the purpose of an amateur radio licence?

A. Broadcasting advertisements for a paid service
B. Allocating paid taxi journeys
C. Investigating how a different antenna affects a hobby radio contact
D. Running a commercial dispatch network

2. Which sequence names the three UK amateur licence levels in the usual progression?

A. Foundation → Intermediate → Full
B. Foundation → Professional → Full
C. Beginner → Foundation → Commercial
D. Intermediate → Foundation → Full

3. A signal has a frequency of 14.2 MHz. Which broad spectrum range contains it?

A. VHF
B. HF
C. UHF
D. None of these

4. A frequency of 145 MHz means:

A. A wavelength of 145 metres
B. 145 cycles per minute
C. 145,000 cycles per second
D. 145,000,000 cycles per second

5. Which description defines wavelength?

A. The number of complete cycles in one second
B. The distance occupied by one complete spatial cycle
C. The number of messages sent in one minute
D. The strength of the received audio

6. The frequency doubles and the wave speed stays constant. What happens to wavelength?

A. It doubles.
B. It stays the same.
C. It halves.
D. It becomes four times as long.

7. Which broad range contains 435 MHz?

A. UHF
B. HF
C. VHF
D. The audio range

8. Use the EX307 conversion chart. Which wavelength is closest to that of a 145 MHz signal?

A. 20 m
B. 200 m
C. 2 m
D. 0.02 m

9. Which term describes varying the instantaneous frequency of a carrier to convey information?

A. AM
B. SSTV
C. CW
D. FM

10. A handheld radio can tune from 136 to 174 MHz. What does that tell you about your permission to transmit?

A. Every frequency in that range is available under a Foundation licence.
B. It describes the equipment's capability; permission must be checked separately.
C. Any frequency is permitted if nobody answers there.
D. Frequencies are permitted whenever FM is selected.

11. What does secondary allocation status mean for an amateur service sharing with a primary service?

A. It must avoid harmful interference to the primary service and cannot demand protection from that service's legitimate operation.
B. It has exclusive use after dark.
C. It can require the primary service to move frequency.
D. It means only beginners may use the band.

12. Read the 144–146 MHz row in the current EX307 Foundation table. What is the maximum peak envelope power listed for ordinary terrestrial operation?

A. 10 W
B. 100 W
C. 1,000 W
D. 25 W

Record your choices

Question	1	2	3	4	5	6	7	8	9	10	11	12
Answer												

Mark with the explained answer key after finishing. If you were unsure, note the uncertainty even when you guessed correctly.

Week 1 explained answers

Try the worksheet and quiz before reading this file.

Quiz

Question	Answer	Explanation
1	C	Comparing antennas for personal learning fits the hobby's purpose. An amateur licence is not permission for commercial dispatch or advertising.
2	A	The three levels are Foundation, Intermediate and Full.
3	B	14.2 MHz lies between 3 and 30 MHz, so it is HF.
4	D	Mega means one million. Hz means cycles per second.
5	B	Wavelength is a distance. Frequency counts cycles in time.
6	C	At constant speed, frequency and wavelength vary inversely: double one and halve the other.
7	A	UHF spans 300–3,000 MHz. 435 MHz falls inside it.
8	C	The chart gives about 2 m. An optional calculation gives about 2.07 m. The question asks for the closest estimate.
9	D	FM stands for frequency modulation. AM changes amplitude; CW keys a carrier; SSTV describes an image mode.
10	B	A tuning range is a hardware capability. The licence, relevant limits and operating conditions determine permission.
11	A	Secondary status does not give priority over a primary service. It is unrelated to the operator's licence level or experience.
12	D	The referenced row lists 25 W for ordinary terrestrial operation. Do not apply one row's limit to every band or type of operation.

Question 12 was checked against EX307 dated 22 January 2026 and Ofcom Schedule 1, Table A. Refresh it if the official reference changes.

Worksheet

Activity A

1. **Fits the purpose:** comparing antennas is technical learning.
2. **Does not fit:** allocating paid jobs is business traffic.
3. **Fits the purpose:** exchanging reception observations is a hobby/technical activity.

These classifications answer the purpose question only. Other licence conditions still apply to an actual transmission.

Activity B

Given frequency	Frequency in MHz	Range
7.1 MHz	7.1	HF
14.2 MHz	14.2	HF
100 MHz	100	VHF
145 MHz	145	VHF
435 MHz	435	UHF
2.4 GHz	2,400	UHF

VHF describes a broad spectrum range. Only specific parts are amateur allocations; a 100 MHz example does not become an amateur channel because it is VHF.

Activity C

1. **6 Hz.** Count complete cycles in the stated second.
2. **Wavelength**, because the axis represents distance.
3. **Halves.** Wave speed was held constant.
4. About **21 m** near 14 MHz and about **2 m** near 145 MHz. Accept roughly 20–22 m and 1.9–2.2 m for this classroom chart-reading task; these are RARS teaching tolerances, not official marking rules.
5. About **150 MHz** from exactly 2 m. A coarse graph reading near this is reasonable. It is not an instruction to transmit at 150 MHz.
6. Band names are convenient approximations; frequencies vary across the band, so their wavelengths vary too.

Optional numerical check: 300 ÷ 145 ≈ **2.07 m**.

Activity D

Description	Term
Keyed carrier for Morse	CW
Speech represented digitally	Digital voice
Still picture transmitted slowly	SSTV
Instantaneous frequency varied by information	FM
One sideband, carrier suppressed	SSB

The radio changed **mode**. Its displayed frequency and band did not change.

Activity E

Item	Expected lookup
Lower and upper limits	144 and 146 MHz
Amateur service status	Primary
Ordinary terrestrial maximum peak envelope power	25 W
Amateur service status at 10,100–10,150 kHz	Secondary

Frequency alone leaves out power, status, special restrictions, mode/bandwidth suitability and other operating conditions. Read the complete relevant row and applicable licence conditions.

Choose what to revisit

If you found this difficult…	Try this next
HF/VHF/UHF classification	Write the four boundaries, then place three new examples between them.
Units	Convert 1.5 MHz into Hz, then explain what the number counts. Answer: 1,500,000 cycles per second.
Wavelength relationship	Sketch two waves over the same distance and explain which has the higher frequency at equal speed.
Chart reading	Work in both directions and label the units aloud.
Modes	Explain the difference between choosing a frequency and choosing FM.
Table reading	Point to each column as you explain what it contributes to a decision.

RARS learning target: 10/12 plus an explanation of corrections. The official exam uses a different length and covers the full Foundation syllabus; this week's score is not an exam-readiness assessment.

Week 1 tutor guide

Session: Your first steps in amateur radio · **Duration:** 90 minutes · **Group:** beginners

Use the learner guide as the explanatory text. These notes support a discussion-led class, not a verbatim lecture.

Prepare the room

- Open or print the current EX307 booklet, the worksheet and quiz. Keep the answer key for the review.
- Have a whiteboard, a metre rule or tape measure and a projector if available.
- Display the two original SVG diagrams. They also work as ordinary printed figures.
- An optional receiver/SDR can make the subject tangible. The lesson does not depend on live reception or internet access once the resources are downloaded.
- If showing a transceiver, prevent accidental transmission during this introductory session. Use receive-only equipment or a suitable transmit-inhibit arrangement. Any separate on-air demonstration must be prepared and directly supervised under the applicable licence conditions.

Explain that learners can ask for a question to be rephrased. Introduce one acronym at a time, let people work in pairs, and give them time to read numbers before answering aloud. Do not rely on colour to identify bands or waves.

Run the session

Minutes	Segment	What the tutor does	Evidence of learning
0–8	Start with a purpose	Ask what a radio could do on campus. Introduce frequency, band and mode using the display example.	Learners offer a use and identify a question they need answered.
8–18	Amateur radio and licensing	Discuss hobby versus business use; introduce the three levels and the exam/licence distinction.	Pairs explain worksheet Activity A.
18–32	A shared spectrum	Teach the 3–30–300–3,000 sequence. Sort the frequency cards in Activity B.	Each learner classifies one example and explains why tuning capability is not permission.
32–48	Waves and units	Compare the two wave plots; measure a metre; practise the official chart with Activity C.	Learners state both units and predict what happens when frequency doubles.
48–53	Short break	Pause and invite quiet questions.	—
53–63	Modes	Explain the mode table; run Activity D as pair matching.	Learners distinguish a mode change from a frequency change.
63–73	Read the reference table	Work through Activity E, including the whole 144–146 MHz row.	Learners find status and power, and notice units and qualifications.
73–85	Individual quiz	Allow about 8 minutes to answer and 4 to compare selected reasoning. Permit EX307 for Questions 8 and 12.	Collect uncertain topics as well as scores.
85–90	Exit check and homework	Ask the three exit questions below; explain the 30-minute practice task.	Each learner identifies one idea they can explain and one to revisit.

If reading the reference booklet takes longer, finish reviewing the quiz as homework. Preserve the discussion of misunderstandings instead of rushing into extra terminology.

Explanations that usually help

Why a licence? A transmitter can affect people who are nowhere near the room. Permission and operating skills help us share the resource responsibly. Keep geographical restrictions, call-sign details and licence administration for Week 2.

Frequency versus wavelength: draw the same pattern with one horizontal axis labelled time and a separate horizontal axis labelled distance. Ask what each axis lets us measure. In the frequency illustration, count complete cycles, not just the number of visible peaks.

Approximate band names: use 145 MHz ≈ 2.07 m and 14.2 MHz ≈ 21.1 m. Explain that the names “2 metres” and “20 metres” are conventions. If someone calculates 20 m as 15 MHz, their physics calculation is right; 15 MHz is not thereby an amateur allocation.

Chart reading: model one frequency-to-wavelength example, then one in the reverse direction. Explain logarithmic spacing. Accept estimates appropriate to the printed graph, rather than demanding calculator precision.

Mode vocabulary: AM, FM and SSB concern ways of modulating a signal. Data modes and SSTV describe wider ways of carrying messages or pictures. An introductory list does not make them

mutually exclusive technical categories. Students need recognition now; waveform detail returns in Week 4.

Sharing: primary does not mean sole owner. Secondary operation must protect primary services. Do not imply that an apparently silent frequency is automatically available to transmit on.

Scope and assessment

The source map identifies partial coverage. In particular, this week does not complete licence conditions, AC theory, waveform recognition or operating procedure.

The quiz is formative. As a RARS teaching target, aim for **10/12 with an explanation of corrected mistakes**. This is not an official pass mark and does not predict the Foundation exam result. Revisit Questions 1, 10 and 11 even when the total score is high: the reasoning matters for responsible use.

Use the answer key to choose a focused follow-up. A learner who remembers “2 metres” but cannot explain the relationship needs a different activity from a learner who understands the idea but misreads MHz as kHz.

Exit questions

1. What changes when we change the frequency, and what changes when we change the mode?
2. If frequency increases, what happens to wavelength?
3. Where would you check whether a transmission is permitted?

Homework: about 30 minutes

- 10 minutes: reread the sections connected to any quiz mistakes.
- 10 minutes: repeat the chart and table lookups without following the worked answers.
- 5 minutes: make three flashcards for the HF/VHF/UHF boundaries.
- 5 minutes: prepare a one-minute explanation of a radio activity you would like to try.

Optional listening is enrichment, not a requirement. If you add it, arrange an amateur-band receive-only demonstration and have a fallback explanation when no signal is present.

Sources, scope and editorial notes

Checked: 6 October 2026. **Material version:** 0.1.

Primary references

ID	Reference	What it was used to verify
S1	RSGB combined syllabus v1.6b, published 5 March 2026	Foundation-column learning points and scope; distinction between exam and live band plans
S1a	March 2026 change summary	Latest minor edits to the linked specification
S2	EX307 Foundation reference booklet, 22 January 2026	Page 2 parameter table; page 4 conversion chart; result-to-licence process
S3	RSGB specification landing page	Current document links and exam overview
S4	Ofcom Amateur Radio Wireless Telegraphy Licence Conditions, OFW611	Condition 6: purpose, interference and supervision; Schedule 1 Table A: Foundation allocations and power
S5	Ofcom amateur licensing information	Current entry point for conditions, guidance and coordination documents

EX307 is also linked from the RSGB exam forms page. Use the landing pages to check for replacements; do not assume a dated PDF remains current indefinitely.

Foundation syllabus mapping

The identifiers below refer specifically to the **Foundation column**. “Partial” means the learning point also contains material deferred to later weeks.

Point	Week 1 treatment	Follow-up
1A1	Partial: purpose and non-commercial use	Geographical and airborne/ship conditions
1A2	Partial: licence levels	Locators, suffixes and associated restrictions
1B1	Introduction: supervision	Callsign use and complete operating context
1D1	Partial: interference responsibility	Inspection, restrictions and closedown
1H1	Practise status and two table rows	Further rows, restrictions and table exercises
2E2	Partial: frequency ranges and wave vocabulary	AC/AF/RF details and other specified frequencies
2E7	Relationship, units and chart practice	Reinforce with antennas
3A1–3A3	Introduction: modes and basic modulation	Full treatment of modulation and bandwidth

The optional 300/f calculation supports intuition. The Foundation 2E7 column does not specify the formula-manipulation requirement found in the Intermediate column. This module is not a complete syllabus coverage claim.

Transcript provenance and original authorship

The course organiser supplied a transcript described as Essex Ham's introductory Foundation module on 6 October 2026. It supplied the starting topics: the hobby, spectrum sharing, frequency/wavelength and modes. The exact video URL, publication date and version were not supplied, so none has been invented.

Acknowledgement: **Essex Ham / Foundation Online**, whose introductory teaching inspired the topic selection. The learner text, campus scenario, exercises, quiz and diagrams in this pack were newly authored for RARS. This repository does not reproduce the supplied transcript, video stills, slides or logos. External reference documents are linked rather than bundled. There is no claim of affiliation or endorsement.

Changes made when developing the lesson

- Replaced the spoken-video format and promotions with learning outcomes, a classroom sequence and written activities.
- Corrected transcription errors such as “Anor/amer” to amateur, “MOS” to Morse and “mahz” to MHz.
- Corrected “millions of hertz per second”: MHz means millions of cycles per second.
- Distinguished time-axis frequency/period explanations from distance-axis wavelength diagrams.
- Treated wavelength band names as approximate. At 145 MHz the wavelength is about 2.07 m; at 14.2 MHz it is about 21.1 m. Exactly 20 m gives about 15 MHz, not an amateur 20-metre operating frequency.
- Used the current EX307 contents for exercises. The January 2026 booklet has the Foundation parameter table, specimen band plans and conversion chart; the lesson does not promise the separate general VHF allocation table mentioned in the transcript.
- Replaced unsupported or nonessential historical claims and operator-number rankings with practical beginner examples. These are not needed for this module's outcomes.
- Clarified that FT8 uses structured data exchanges and SSTV typically sends still pictures.
- Kept formula manipulation as an optional aid and identified incomplete syllabus coverage.

Figures and questions

assets/spectrum.svg and assets/waves.svg are original vector illustrations made for this lesson. They contain no third-party photographs or copied diagrams. The spectrum figure shows broad ranges and three illustrative points, not a complete allocation map. The wave figure uses equal spatial spans and equal propagation speed to explain the inverse relationship.

All 12 quiz questions are newly written practice questions. They are not copied from or presented as RSGB exam-bank questions. The answer key records the current reference-table values and explains where a lookup is required.

Next review

Before using this draft with a cohort, the tutor should work through the exercises with the current official references and adjust pacing for the group. Before expanding the course, build the remaining Foundation coverage map and revisit every partial point above.